

G.H. RAISONI COLLEGE OF ENGINEERING AND MANAGEMENT, WAGHOLI, PUNE

An Empowered Autonomous Institute, Affiliated to SPPU, Pune
NAACA credited A+ Grade

Department of CSE (Data Science)

TAE-2

Fake Email Detection and Management System

Guided by:

Mr. Chinmay Mukin

Submitted by:-

Abhinav Komalwar(P-02)



Project Overview



What is a fake email?

A message that pretends to be from a trusted sender to trick users or steal data.



Why detection?

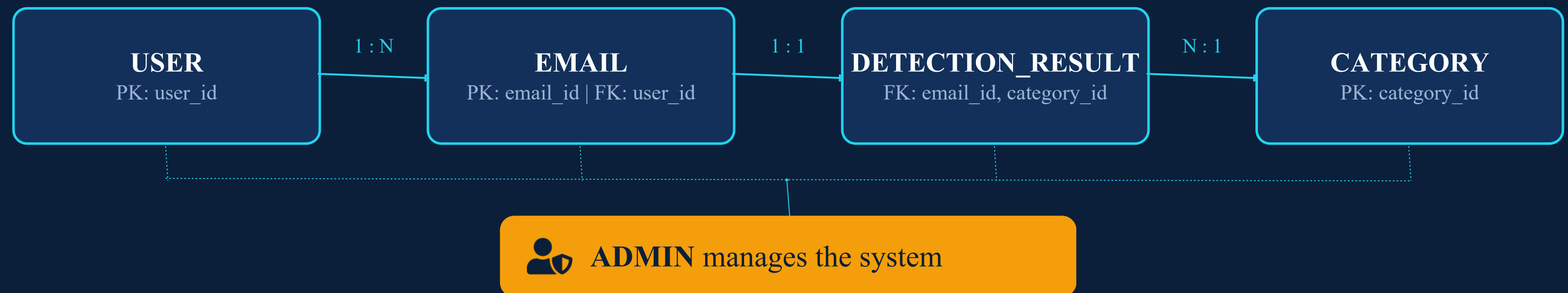
Stops phishing, spam and fraud, and keeps user data and accounts safe.



Purpose of the system

Store, classify and manage emails and their detection results using tables.

The database stores each email, its detection result and category in linked tables





Database Design and JOIN Operations

Table	Key attributes
USER	user_id, name, email_address
EMAIL	email_id, user_id, sender_email, subject, message, received_date
DETECTION_RESULT	result_id, email_id, category_id, confidence_score, detection_status
CATEGORY	category_id, category_name

SQL JOIN QUERY

```
SELECT
    u.name,
    e.sender_email,
    e.subject,
    d.detection_status,
    d.confidence_score,
    c.category_name
FROM USER u
JOIN EMAIL e ON u.user_id = e.user_id
JOIN DETECTION_RESULT d ON e.email_id = d.email_id
JOIN CATEGORY c ON d.category_id = c.category_id;
```



JOIN combines data from multiple related tables.

Each JOIN links two tables through a common key.

USER ↔ EMAIL
user_id

EMAIL ↔ RESULT
email_id

RESULT ↔ CATEGORY
category_id



Storing the Joined Dataset

The JOIN result can be saved as a separate table for easier analysis and reporting.

```
CREATE TABLE ... AS SELECT
```

```
CREATE TABLE Fake_Email_Report AS  
SELECT
```

```
    u.name,  
    e.sender_email,  
    e.subject,  
    d.detection_status,  
    d.confidence_score,  
    c.category_name
```

```
FROM USER u
```

```
JOIN EMAIL e ON u.user_id = e.user_id
```

```
JOIN DETECTION_RESULT d ON e.email_id = d.email_id
```

```
JOIN CATEGORY c ON d.category_id = c.category_id;
```



Fake_Email_Report (stored table)

Name	Sender Email	Subject	Status	Category	Confidence
Rahul	abc@mail.com	Win Prize	Fake	Phishing	95%
Priya	xyz@gmail.com	Meeting	Genuine	Normal	98%
Amit	offer@xyz.com	Free Gift	Fake	Spam	91%



Stored once, then reused for queries, reports and views.

JOIN of 4 tables

CREATE TABLE ... AS

Fake_Email_Report



Creating Views for Fake Email Management



What is a VIEW?

A virtual table based on an SQL query. It stores the query, not a copy of the data.

Benefits



Easy reporting



Faster access to information



Better data organization



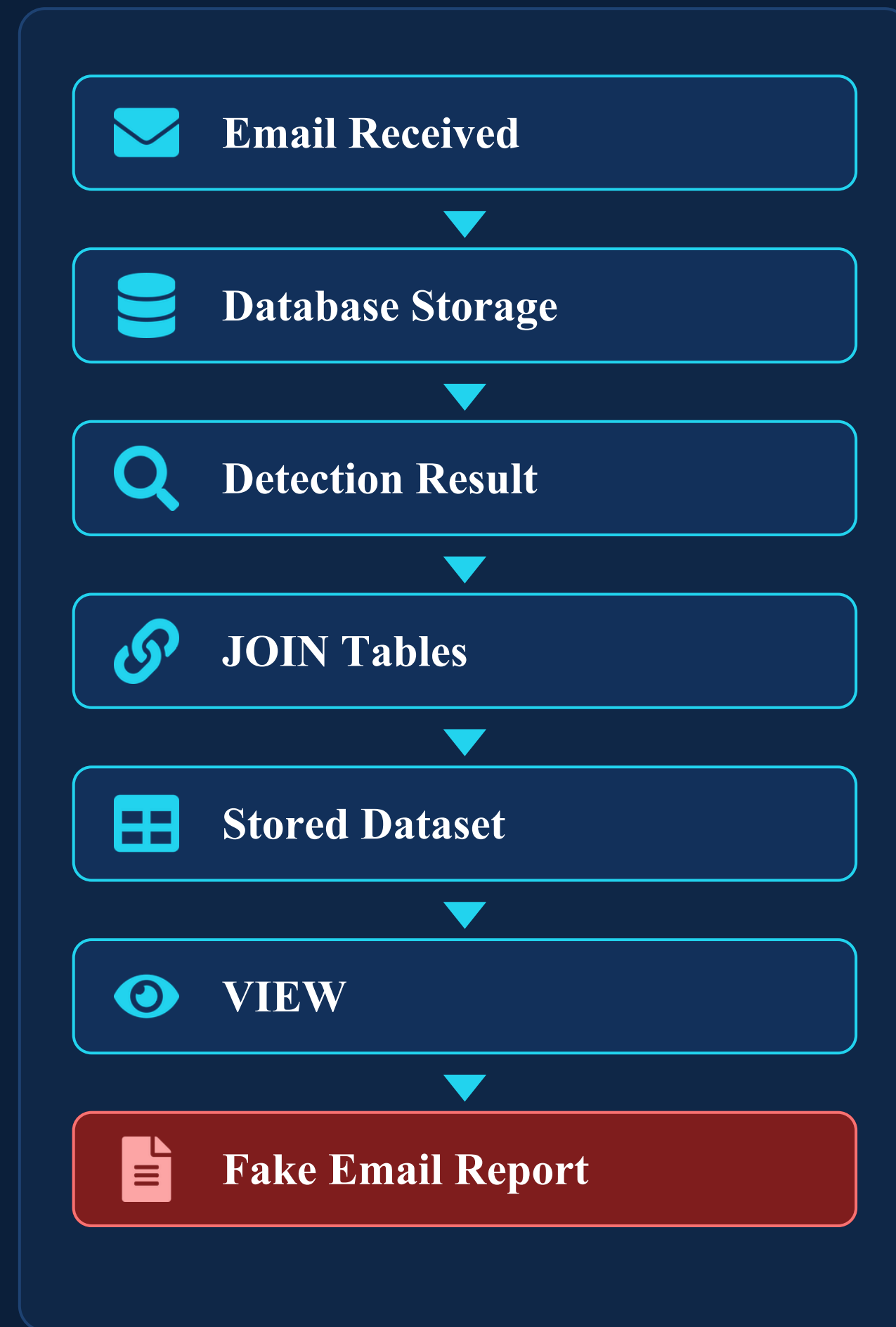
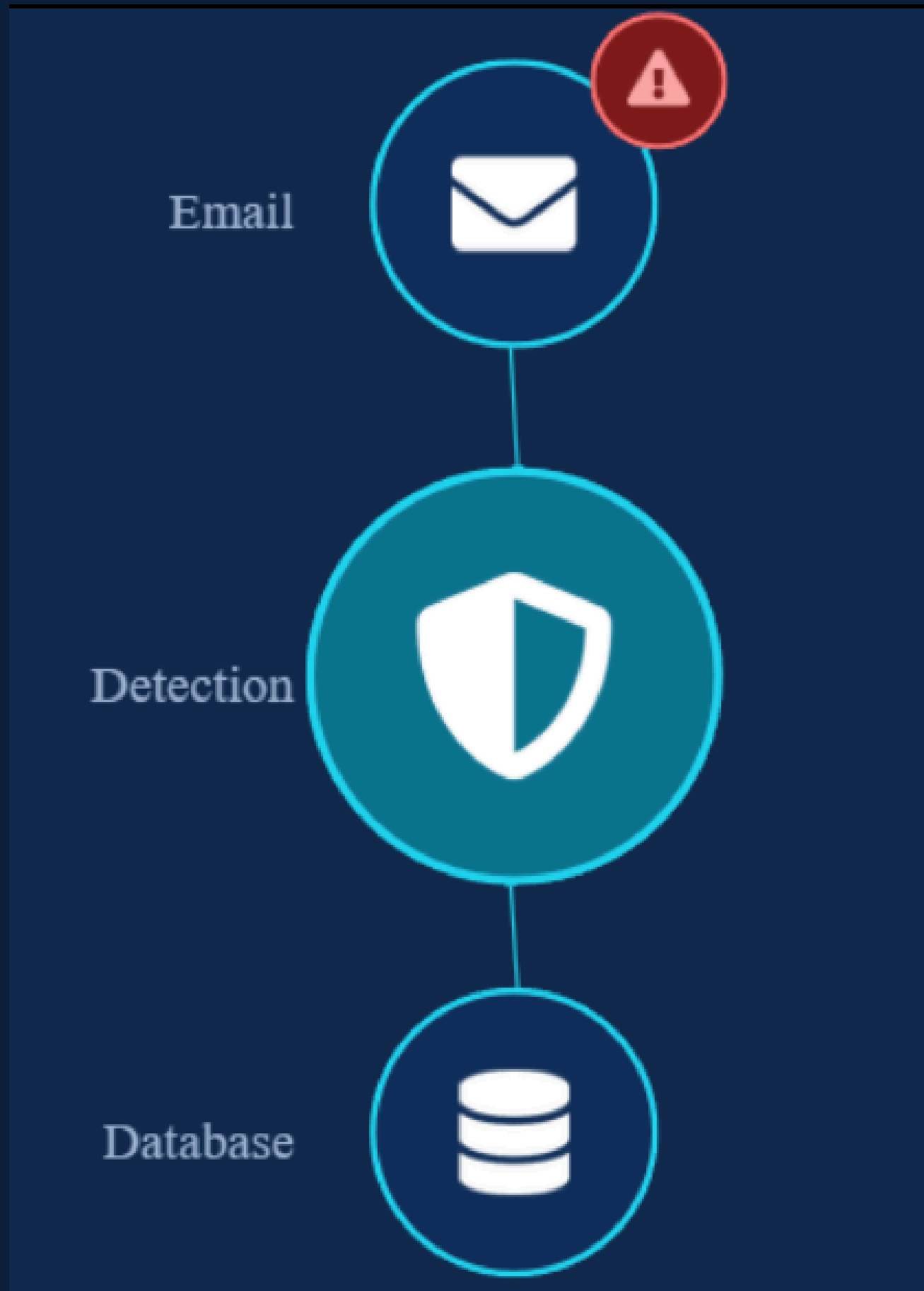
Useful for administrators

CREATE VIEW

```
CREATE VIEW Fake_Email_View AS
SELECT
    sender_email,
    subject,
    category_name,
    detection_status,
    confidence_score
FROM Fake_Email_Report
WHERE detection_status = 'Fake';
```

Admin dashboard output `SELECT * FROM Fake_Email_View;`

Sender Email	Subject	Category	Status	Confidence
abc@mail.com	Win Prize	Phishing	Fake	95%
offer@xyz.com	Free Gift	Spam	Fake	91%



Thank you !